Sensor Calibration for Urban Air Quality Monitoring: A Regression Analysis of Metal Oxide Sensor Responses

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Course: DATA 210P: Statistical Methods I

Dataset: Air Quality UCI – UCI Machine Learning Repository (ID: 360)

Statistical Methods: Simple Linear Regression, Multiple Linear Regression, Ridge Regression, Lasso Regression, Assumption Diagnostics

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1 Abstract

Accurate monitoring of urban air quality is critical for public health, yet certified reference analyzers are costly and difficult to deploy at scale. Metal oxide chemical sensors offer a low-cost alternative, but their responses are affected by cross-sensitivity, sensor drift, and environmental conditions, making calibration a key challenge. This report investigates whether ground truth pollutant concentrations and environmental variables can reliably predict the tin oxide sensor response PT08.S1(CO) using one year of hourly measurements collected at road level in an Italian city, sourced from the Air Quality UCI dataset (ID: 360), which contains 9,357 observations across 12 variables after preprocessing.

The analysis follows a full regression pipeline. Exploratory data analysis identified missing value patterns, skewed distributions, and strong multicollinearity among predictors, with a correlation as high as 0.886 between C6H6(GT) and PT08.S1(CO). Four models were fitted on an 80/20 time-ordered train/test split: Simple Linear Regression (SLR) as a baseline, Multiple Linear Regression (MLR) using all 7 predictors, and Ridge and Lasso regression with 5-fold cross-validation to select the optimal regularization parameter λ. Assumption diagnostics were applied to the MLR model, assessing linearity, homoscedasticity, normality, and influential observations.

Results show a clear improvement as model complexity increases. SLR achieved a test R^2 of 0.5835, while MLR improved this to 0.8229. Ridge and Lasso further improved generalization through regularization, with Lasso achieving the best test R^2 of 0.8462 and lowest test RMSE of 80.19 using only 4 active predictors – C6H6(GT), NOx(GT), NO2(GT), and RH. Lasso is selected as the final model for its superior predictive accuracy, automatic variable selection, and ability to address the strong multicollinearity identified in diagnostics. However, heteroscedasticity and heavy-tailed residuals remain limitations, suggesting that non-linear models may offer further improvement.

2 Introduction

Air pollution is one of the most pressing environmental health challenges in modern urban societies. Prolonged exposure to pollutants such as Carbon Monoxide (CO), Benzene (C6H6), Nitrogen Oxides (NOx), and Nitrogen Dioxide (NO2) has been linked to respiratory disease, cardiovascular conditions, and premature mortality. Accurate and continuous monitoring of these pollutants is therefore essential for informing public health policy and urban planning. However, certified reference analyzers – the gold standard for air quality measurement – are expensive, require specialist maintenance, and are impractical to deploy at scale across a city. Metal oxide chemical sensors offer a promising low-cost alternative, but their responses are known to be affected by cross-sensitivity to other pollutants, sensor drift over time, and environmental conditions such as temperature and humidity, making reliable calibration a significant technical challenge.

This report uses the Air Quality UCI dataset (UCI Machine Learning Repository, ID: 360), which contains 9,357 hourly averaged measurements collected from a multisensor device deployed at road level in a significantly polluted Italian city between March 2004 and February 2005. The dataset includes responses from five metal oxide chemical sensors alongside ground truth pollutant concentrations from a co-located certified reference analyzer, and environmental variables including temperature, relative humidity, and absolute humidity. After removing NMHC(GT) due to 90.23% missingness, the final dataset retains 7 predictors and PT08.S1(CO) – the tin oxide sensor nominally targeting CO – as the response variable.

The aim of this report is to investigate whether ground truth pollutant concentrations and environmental variables can reliably predict the PT08.S1(CO) sensor response using linear regression methods. Specifically, the goals are: (1) to perform exploratory data analysis to understand the structure, distributions, and relationships in the data; (2) to fit and evaluate four regression models – SLR, MLR, Ridge, and Lasso – of increasing complexity; (3) to assess whether standard linear regression assumptions hold through diagnostic plots; and (4) to select the most suitable model based on predictive accuracy, interpretability, and robustness to multicollinearity. The findings contribute to the broader goal of developing reliable low-cost sensor calibration methods for urban air quality monitoring.

3 Review

Urban air quality monitoring has become an increasingly active research area as cities seek cost-effective alternatives to expensive certified reference analyzers. Metal oxide chemical sensors have been widely studied as low-cost substitutes, but their practical deployment is complicated by well-documented issues of cross-sensitivity, sensor drift, and sensitivity to environmental conditions such as temperature and humidity. De Vito et al. (2008) first demonstrated the use of this dataset to investigate these challenges, showing that sensor responses are influenced not only by their target pollutant but also by other co-present gases, making direct sensor readings unreliable without calibration. Subsequent studies have confirmed that regression-based calibration models – which use ground truth reference measurements alongside environmental variables to predict sensor responses – can substantially improve sensor accuracy in field deployments. The Air Quality UCI dataset used in this report has since become a standard benchmark for evaluating such calibration approaches, given that it represents the longest freely available recording of on-field deployed air quality chemical sensor responses.

From a statistical perspective, the high multicollinearity among air quality variables – arising from the physical co-variation of pollutants in urban environments – poses a known challenge for Ordinary Least Squares regression. Ridge regression, introduced by Hoerl and Kennard (1970), addresses this by adding an L2 penalty to shrink correlated coefficients toward zero, stabilizing estimates without eliminating predictors. Lasso regression, proposed by Tibshirani (1996), extends this with an L1 penalty that performs automatic variable selection by shrinking some coefficients exactly to zero, producing sparser and more interpretable models. Both methods have been widely applied in environmental and atmospheric science settings where predictor multicollinearity is common, and are well-suited to the air quality calibration problem where identifying the most influential pollutants is as important as predictive accuracy.

4 Libraries & packages

```
# Standard library imports
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import tabulate
from IPython.display import display, Math

# Third-party imports
```

```python
from dataclasses import dataclass
from typing import Callable, Dict, Optional, Tuple, List, Any
from textwrap import dedent
import patsy
import textwrap
import pymc as pm
import arviz as az

# Statsmodels imports
import statsmodels.api as sm
import statsmodels.formula.api as smf
from statsmodels.stats.outliers_influence import variance_inflation_factor
from statsmodels.graphics.gofplots import qqplot
from statsmodels.nonparametric.smoothers_lowess import lowess
from statsmodels.stats.diagnostic import het_breuschpagan
from statsmodels.genmod.generalized_linear_model import SET_USE_BIC_LLF
from statsmodels.tools.sm_exceptions import PerfectSeparationError

# Sklearn imports
from sklearn.model_selection import KFold
from sklearn.metrics import mean_squared_error
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression, RidgeCV, LassoCV
from sklearn.linear_model import Ridge, Lasso, lasso_path
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import StratifiedKFold, cross_val_predict
from sklearn.metrics import roc_auc_score, RocCurveDisplay
from sklearn.linear_model import LogisticRegressionCV
from sklearn.utils import resample
from sklearn.metrics import r2_score, mean_squared_error

# Set global configurations
np.random.seed(42)
pd.set_option("display.max_columns", 200)
pd.set_option("display.width", 120)
```

5 Air Quality Data Dictionary:

Source: UCI Machine Learning Repository (ID: 360)
Records: 9,357 hourly observations | **Period:** March 2004 - February 2005

Missing values: Tagged as `-200` in the original data

Index Variables

1. **Date**
   - Date of recording
   - Type: Date (DD/MM/YYYY)
   - Modeling note: Combined with Time to create Datetime index
2. **Time**
   - Hour of recording
   - Type: Categorical (HH.MM.SS)
   - Modeling note: Combined with Date to create Datetime index

---

Response Variables – Sensor Readings

3. **PT08.S1(CO)**
   - Hourly averaged tin oxide sensor response, nominally CO targeted
   - Type: Categorical (Continuous sensor output)
   - Units: –
   - Modeling note: Response variable 1; measures sensor estimation capability for CO; 3.9% missing (-200)
4. **PT08.S2(NMHC)**
   - Hourly averaged titania sensor response, nominally NMHC targeted
   - Type: Categorical (Continuous sensor output)
   - Units: –
   - Modeling note: Response variable 2; measures sensor estimation capability for NMHC; 3.9% missing (-200)
5. **PT08.S3(NOx)**
   - Hourly averaged tungsten oxide sensor response, nominally NOx targeted
   - Type: Categorical (Continuous sensor output)
   - Units: –
   - Modeling note: Response variable 3; measures sensor estimation capability for NOx; 3.9% missing (-200)

6. **PT08.S4(NO2)**
   - Hourly averaged tungsten oxide sensor response, nominally NO2 targeted
   - Type: Categorical (Continuous sensor output)
   - Units: –
   - Modeling note: Response variable 4; measures sensor estimation capability for NO2; 3.9% missing (-200)
7. **PT08.S5(O3)**
   - Hourly averaged indium oxide sensor response, nominally O3 targeted
   - Type: Categorical (Continuous sensor output)
   - Units: –
   - Modeling note: Response variable 5; measures sensor estimation capability for O3; 3.9% missing (-200)

Predictor Variables – True Pollutant Concentrations (Reference Analyzer)

8. **CO(GT)**
   - True hourly averaged concentration of Carbon Monoxide
   - Type: Integer (Continuous)
   - Units: mg/m3
   - Modeling note: Ground truth predictor for PT08.S1(CO); 18.0% missing (-200)
9. **NMHC(GT)**
   - True hourly averaged overall Non-Metanic Hydrocarbons concentration
   - Type: Integer (Continuous)
   - Units: microg/m3
   - Modeling note: Ground truth predictor for PT08.S2(NMHC); **90.2% missing** – likely excluded from modeling
10. **C6H6(GT)**
    - True hourly averaged Benzene concentration
    - Type: Continuous
    - Units: microg/m3
    - Modeling note: Ground truth predictor; 3.9% missing (-200)
11. **NOx(GT)**
    - True hourly averaged Total Nitrogen Oxides concentration
    - Type: Integer (Continuous)
    - Units: ppb

- Modeling note: Ground truth predictor for PT08.S3(NOx); 17.5% missing (-200)

12. **NO2(GT)**

    - True hourly averaged Nitrogen Dioxide concentration
    - Type: Integer (Continuous)
    - Units: microg/m3
    - Modeling note: Ground truth predictor for PT08.S4(NO2); 17.6% missing (-200)

Predictor Variables – Environmental Conditions

13. **T**

    - Ambient temperature at time of recording
    - Type: Continuous
    - Units: Celsius (C)
    - Modeling note: Environmental covariate; evidence of cross-sensitivity with sensors; 3.9% missing (-200)

14. **RH**

    - Relative humidity at time of recording
    - Type: Continuous
    - Units: %
    - Modeling note: Environmental covariate; may affect sensor drift; 3.9% missing (-200)

15. **AH**

    - Absolute humidity at time of recording
    - Type: Continuous
    - Units: –
    - Modeling note: Correlated with RH and T; check for multicollinearity before modeling; 3.9% missing (-200)

6 Data Source & Import:

Source: https://archive.ics.uci.edu/dataset/360/air+quality

Load dataset for Air Quality from UC Irvine Machine Learning Repository.

7 Initialization

1. Create target variable
2. Create predictors variable
    - Drop Day and Time
3. Perform data validation
    - Deal with missing value

The dataset is loaded directly from the UCI ML Repository and structured into predictor and response variables. Date and Time columns are dropped as they are not used as predictors in modeling. The response variables y_air consist of the five sensor readings (PT08.S1 through PT08.S5), and the predictor variables X_air contain the remaining 8 columns including ground truth concentrations and environmental variables, giving shapes of (9357, 5) and (9357, 8) respectively. Data validation confirms the full dataset retains 9,357 rows and 13 columns. The presence of -200 values visible in the minimum row of the descriptive statistics confirms that missing value treatment has not yet been applied at this stage.

8 Exploratory data analysis

8.1 1. Check missing values (Missing values are tagged with -200 value)

Missing Value Analysis

Overview The dataset contains a total of **16,701 missing values**, encoded as `-200` in the original data. The missing values fall into three distinct patterns:

Pattern 1 – Structural Blackout (3.91%) All sensor readings and environmental variables (PT08.S1 through PT08.S5, T, RH, AH, C6H6) share exactly **366 missing rows**.

Pattern 2 – Moderate Missingness (~18%) CO(GT), NOx(GT), and NO2(GT) have around 17-18% missing values.

Pattern 3 – Severe Missingness (90.23%) NMHC(GT) has **8,443 out of 9,357 rows** missing – only 9.77% of values are recorded.

Handling Strategy

Step 1 – Drop NMHC(GT) With 90.23% of values missing, NMHC(GT) was removed entirely from the dataset.

Step 2 – Replace -200 with NaN The value -200 is a placeholder tag, not a real measurement. It was converted to NaN so that pandas can recognize it as a missing value.

Step 3 – Forward Fill (ffill) Since this dataset is an **hourly time-series**, each missing value is replaced by the most recent valid observation, preserving the temporal continuity of the data.

After applying these steps, **0 missing values** remain, with the final dataset retaining all 9,357 rows across 7 predictors and 5 response variables.

Table 1: Total missing values in the Air Quality dataset.

	Metric	Value
0	Total missing values (-200)	16701

Table 2: Missing value summary by column.

	missing_count	%_missing
CO(GT)	1683	17.99%
PT08.S1(CO)	366	3.91%
NMHC(GT)	8443	90.23%
C6H6(GT)	366	3.91%
PT08.S2(NMHC)	366	3.91%
NOx(GT)	1639	17.52%
PT08.S3(NOx)	366	3.91%
NO2(GT)	1642	17.55%
PT08.S4(NO2)	366	3.91%
PT08.S5(O3)	366	3.91%
T	366	3.91%
RH	366	3.91%
AH	366	3.91%

8.2 2. Descriptive Statistics

Descriptive Statistics

The predictor variables show large differences in scale and variability. CO(GT) ranges from 0.10 to 11.90 mg/m3 with a mean of 2.08, while NOx(GT) ranges from 2 to 1479 ppb with a mean of 240.73. Among the sensor responses, PT08.S4(NO2) has the highest mean (1453.02) and PT08.S3(NOx) has the widest range (322 to 2683). Environmental variables T and RH show moderate variability, with temperatures ranging from -1.90 to 44.60 Celsius and relative humidity from 9.20 to 88.70%.

Table 3: Descriptive statistics for the Air Quality dataset.

	mean	std	min	max
CO(GT)	2.082	1.470	0.100	11.900
PT08.S1(CO)	1102.730	219.588	647.000	2040.000
C6H6(GT)	10.190	7.566	0.100	63.700
PT08.S2(NMHC)	942.548	269.581	383.000	2214.000
NOx(GT)	240.731	206.618	2.000	1479.000
PT08.S3(NOx)	832.742	255.709	322.000	2683.000
NO2(GT)	109.415	47.222	2.000	340.000
PT08.S4(NO2)	1453.015	347.434	551.000	2775.000
PT08.S5(O3)	1030.512	410.917	221.000	2523.000
T	18.317	8.822	-1.900	44.600
RH	48.817	17.354	9.200	88.700
AH	1.017	0.405	0.185	2.231

8.3 3. Distributions of targets and predictors

Distributions

All five sensor response variables – PT08.S1(CO), PT08.S2(NMHC), PT08.S3(NOx), PT08.S4(NO2), and PT08.S5(O3) – show right-skewed unimodal distributions. The bulk of readings are concentrated in the 500–1500 range, with long right tails indicating occasional high sensor responses. PT08.S3(NOx) and PT08.S5(O3) show the widest spread, while PT08.S1(CO) and PT08.S2(NMHC) are more tightly concentrated around their peaks.

Among the predictor variables, CO(GT), C6H6(GT), and NOx(GT) are all strongly right-skewed, with the majority of values concentrated near zero and long tails extending to the right. NO2(GT) shows a roughly bell-shaped distribution centered around 100 microg/m3. Temperature (T) shows a bimodal distribution, reflecting seasonal variation across the one-year recording period. Relative humidity (RH) is broadly distributed between 10 and 90%, while absolute humidity (AH) shows a bimodal pattern with peaks around 0.5 and 1.0.

8.4 4. Correlation

Correlation Matrix

The correlation matrix reveals strong positive correlations among most sensor responses and pollutant concentrations. PT08.S1(CO) shows strong positive correlations with C6H6(GT) (r = 0.89), PT08.S2(NMHC) (r = 0.90), and PT08.S5(O3) (r = 0.90). PT08.S3(NOx) shows strong negative correlations with most variables, including PT08.S2(NMHC) (r = -0.80) and PT08.S5(O3) (r = -0.80), indicating an inverse relationship. Environmental variables T, RH,

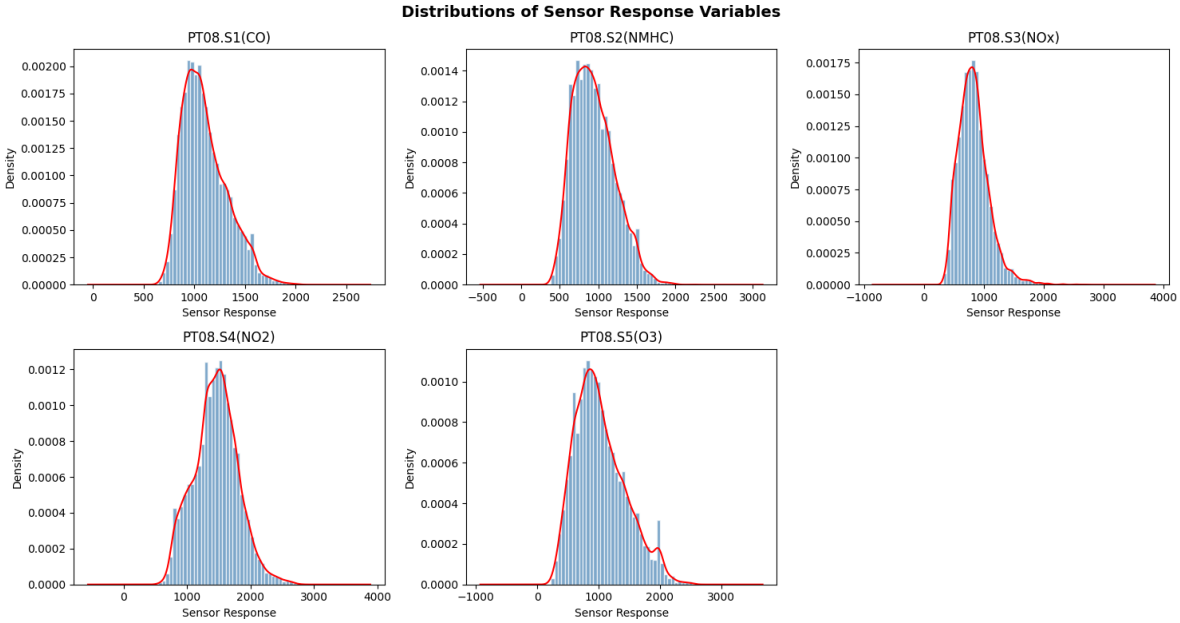

Figure 1: Distributions of sensor response variables.

and AH show weak correlations with most sensor and pollutant variables, with the exception of T and AH (r = 0.65) and T and RH (r = -0.57).

8.5 5. Scatter plots

Scatter Plots – PT08.S1(CO) vs All Variables

PT08.S1(CO) shows the strongest linear relationships with PT08.S2(NMHC) (r = 0.896), PT08.S5(O3) (r = 0.901), and C6H6(GT) (r = 0.886), all displaying clear positive linear trends. The relationship with CO(GT) (r = 0.763) shows a positive trend but with more scatter, including a visible vertical band of points around CO(GT) = 6, which may reflect forward-filled missing values. PT08.S3(NOx) shows a strong negative relationship (r = -0.778) with a curved pattern. Environmental variables T (r = 0.037), RH (r = 0.120), and AH (r = 0.122) show virtually no linear relationship with PT08.S1(CO).

9 Modeling

Based on the EDA, PT08.S1(CO) is selected as the response variable for modeling. Among the five sensor responses, PT08.S1(CO) targets CO – the pollutant with the most complete data after removing NMHC(GT). The scatter plots showed clear linear relationships between

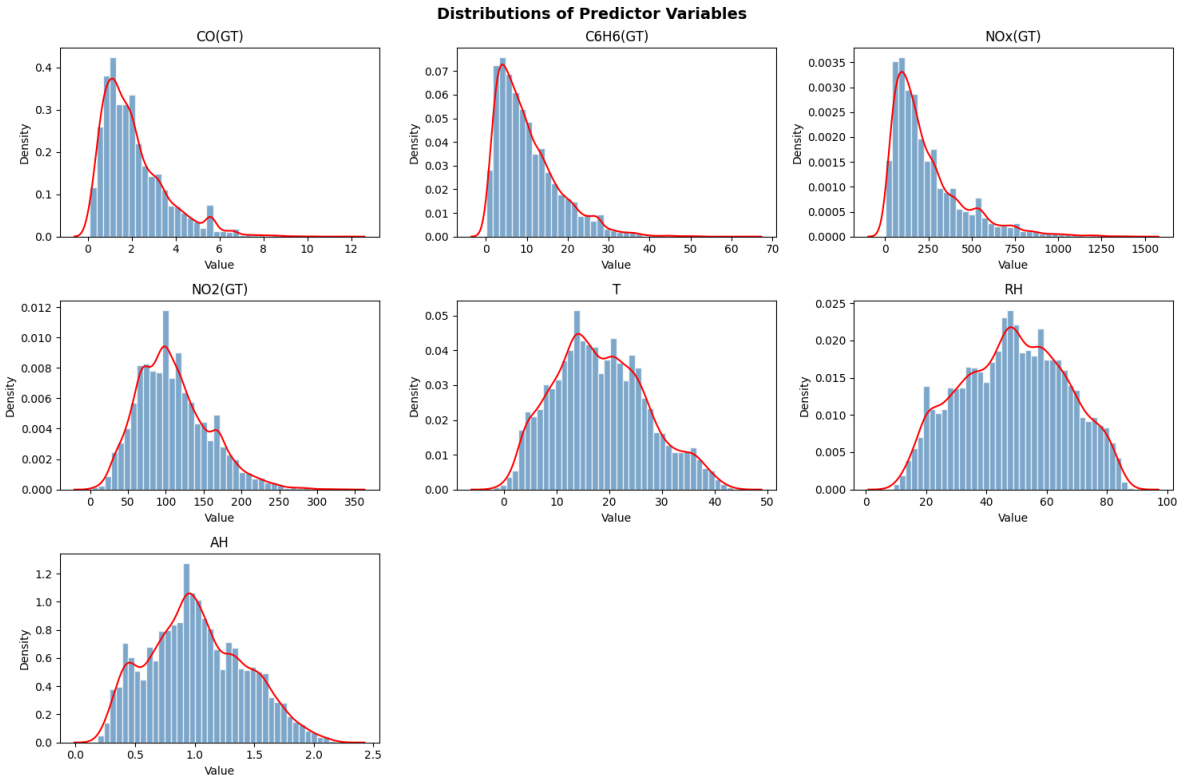

Figure 2: Distributions of predictor variables.

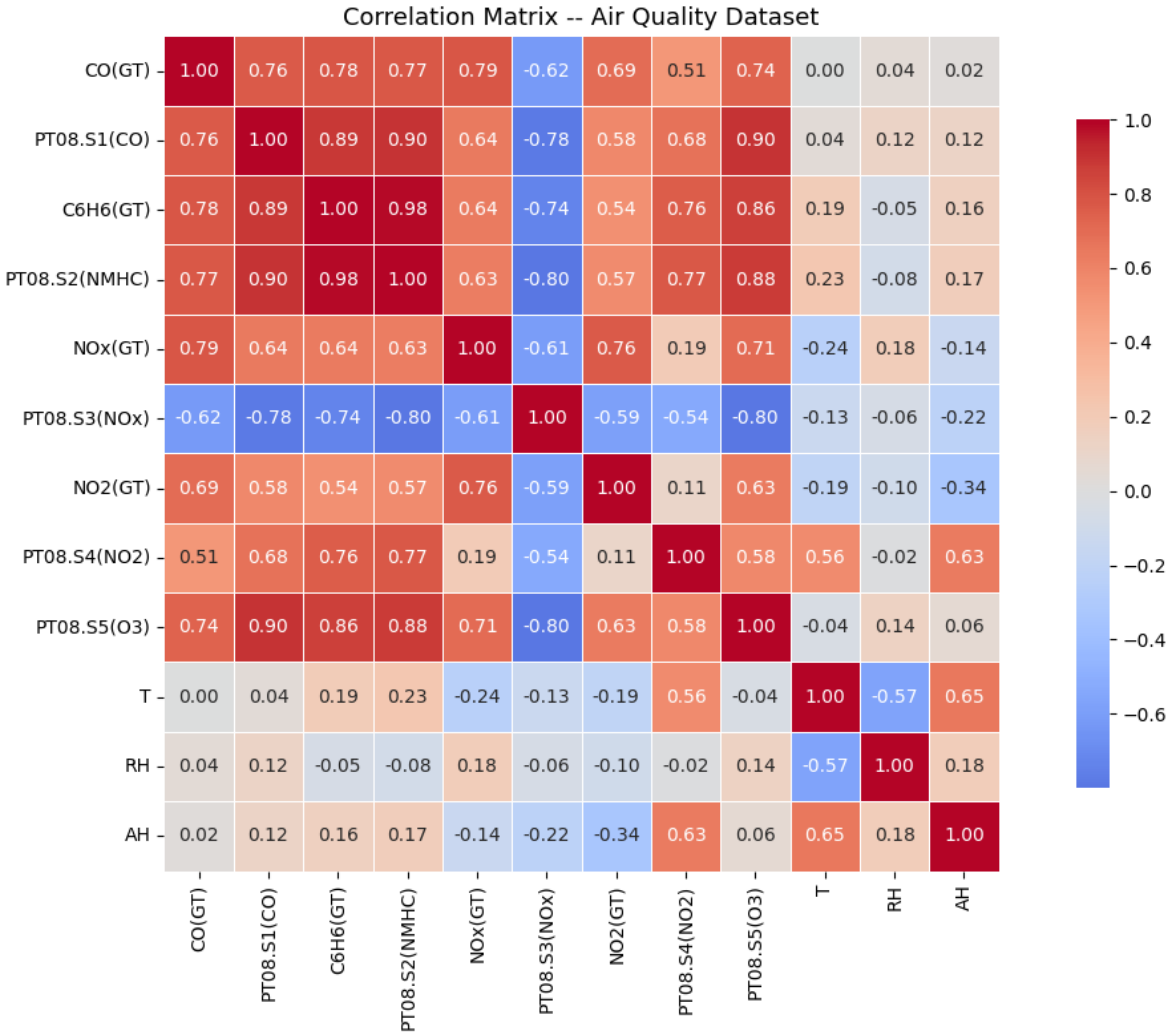

Figure 3: Correlation heatmap for the Air Quality dataset.

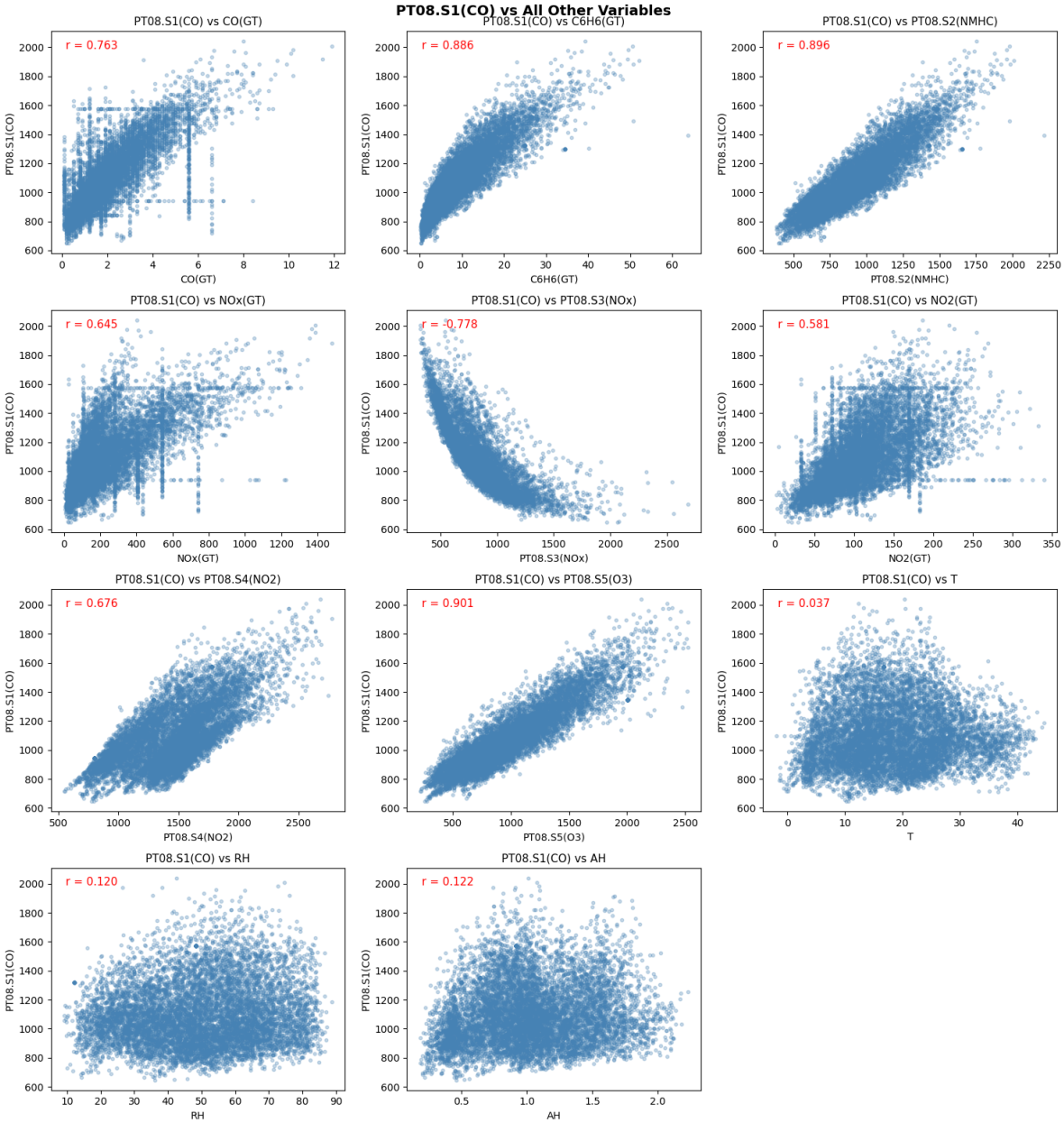

Figure 4: Scatter plots of PT08.S1(CO) vs all other variables.

PT08.S1(CO) and several predictors, particularly C6H6(GT) (r = 0.886), PT08.S2(NMHC) (r = 0.896), and PT08.S5(O3) (r = 0.901), making it well-suited for linear regression analysis. Three models are fitted: Simple Linear Regression (SLR) as a baseline using CO(GT) as the sole predictor, Multiple Linear Regression (MLR) incorporating all predictors to capture combined effects, and Ridge and Lasso regression to address the high multicollinearity observed in the correlation matrix, with Lasso additionally performing variable selection to identify the most influential predictors on sensor estimation capability.

9.1 1. Simple linear regression

$$PT08.S1(CO) = \beta_0 + \beta_1 \cdot CO(GT) + \varepsilon$$

(a) Split the dataset into a training set (80%) and a test set (20%) by time order, then fit the SLR model on the training set.

Table 4: Train/Test split summary for SLR modeling.

		Value
0	Dep. Variable	PT08.S1(CO)
1	No. Observations	7485.0
2	R-squared	0.582
3	Adj. R-squared	0.582
4	F-statistic	10404.612
5	AIC	95669.317
6	BIC	95683.158
7	Log-Likelihood	-47832.658
8	Durbin-Watson	0.208
9	Kurtosis	6.861

(b) Plot the training data (PT08.S1(CO) vs CO(GT)) with the fitted regression line and a 95% confidence band for the mean response.

Figure 5 shows a positive linear relationship between CO(GT) and PT08.S1(CO), with the fitted line increasing from approximately 863 at CO(GT) = 0 to 2200 at CO(GT) = 12. The 95% confidence band is very narrow throughout, reflecting the large training sample size (7,485 observations). However, the spread of points increases as CO(GT) increases, suggesting heteroscedasticity. A vertical band of points is visible around CO(GT) = 6, which corresponds to forward-filled missing values. Overall, the linear trend is clear but the wide scatter around the fitted line is consistent with the R2 of 0.582, indicating that CO(GT) alone explains only 58% of the variation in PT08.S1(CO).

(c) Interpret $\hat{\beta}_1$ in words. No log transformation is applied as variables are in original units.

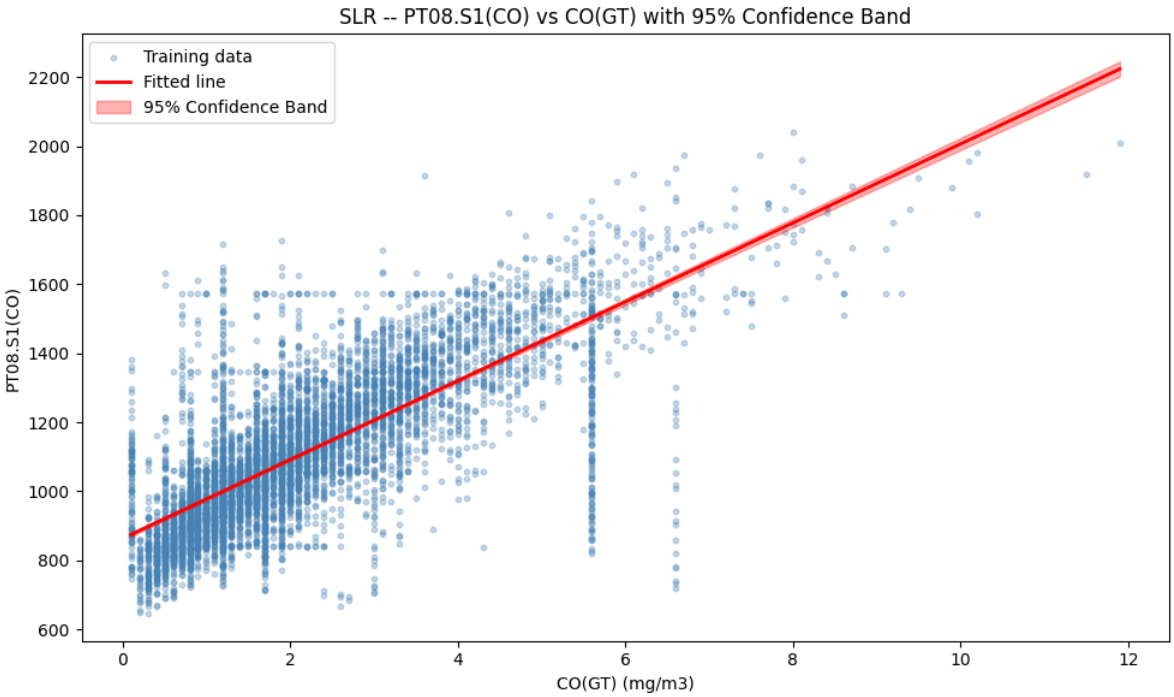

Figure 5: SLR fitted regression line with 95% confidence band for PT08.S1(CO) vs CO(GT).

```
0 (Intercept) = 863.5279
1 (CO(GT))    = 114.2515

Interpretation:
A one-unit increase in CO(GT) (1 mg/m3) is associated with
a 114.2515 unit increase in PT08.S1(CO) sensor response.
No log transformation is applied -- variables are in original units.
```

(d) Evaluate performance on the test set using RMSE and R^2 and comment on generalization.

Table 5 shows that the SLR model achieves an R^2 of 0.5817 on the training set and 0.5835 on the test set, with RMSE of 144.25 and 131.97 respectively. The near-identical train and test performance indicates that the model generalizes well with no signs of overfitting. However, an R^2 of 0.58 suggests that CO(GT) alone explains only 58% of the variation in PT08.S1(CO).

Table 5: SLR train and test performance metrics (RMSE and R2).

	Metric	Train	Test
0	RMSE	144.2474	131.9696
1	R2	0.5817	0.5835

Table 5: SLR train and test performance metrics (RMSE and R2).

Metric	Train	Test

9.2 2. Multiple linear regression

$$PT08.S1(CO) = \beta_0 + \beta_1 \cdot CO(GT) + \beta_2 \cdot C6H6(GT) + \beta_3 \cdot NOx(GT) + \beta_4 \cdot NO2(GT) + \beta_5 \cdot T + \beta_6 \cdot RH + \beta_7 \cdot AH + \varepsilon$$

(a) Fit a full model on the training set using all predictors.

Table 6: MLR model summary for PT08.S1(CO) with all predictors.

		Value
0	Dep. Variable	PT08.S1(CO)
1	No. Observations	7485
2	R-squared	0.839
3	Adj. R-squared	0.839
4	F-statistic	5564.084
5	AIC	88536.551
6	BIC	88591.916
7	Log-Likelihood	-44260.275
8	Durbin-Watson	0.238
9	Kurtosis	5.95
10	Cond. No.	4120.0

(b) Report estimated coefficients, standard errors, and 95% confidence intervals for all predictors.

Table 7: MLR estimated coefficients, standard errors, and 95% confidence intervals.

	Coefficient	Std Error	CI Lower	CI Upper	P-value
Intercept	651.6488	11.9135	628.2950	675.0026	0.0000
CO(GT)	21.6969	1.4767	18.8021	24.5917	0.0000
C6H6(GT)	23.0923	0.2218	22.6576	23.5271	0.0000
NOx(GT)	-0.1939	0.0100	-0.2135	-0.1743	0.0000
NO2(GT)	0.9012	0.0484	0.8064	0.9960	0.0000
T	-0.8041	0.4746	-1.7344	0.1262	0.0902
RH	2.3214	0.1902	1.9485	2.6943	0.0000
AH	13.5528	7.7938	-1.7253	28.8309	0.0821

(c) Interpret $\hat{\beta}$ for all predictors.

From Table 7, the estimated coefficients are interpreted as follows, holding all other predictors constant:

- **Intercept (651.65):** The predicted PT08.S1(CO) sensor response is 651.65 when all predictors are zero.
- **CO(GT) (21.70):** A one-unit increase in CO(GT) (1 mg/m3) is associated with an increase of 21.70 units in PT08.S1(CO).
- **C6H6(GT) (23.09):** A one-unit increase in Benzene concentration (1 microg/m3) is associated with an increase of 23.09 units in PT08.S1(CO).
- **NOx(GT) (-0.19):** A one-unit increase in NOx concentration (1 ppb) is associated with a decrease of 0.19 units in PT08.S1(CO).
- **NO2(GT) (0.90):** A one-unit increase in NO2 concentration (1 microg/m3) is associated with an increase of 0.90 units in PT08.S1(CO).
- **T (-0.80):** A one-unit increase in temperature (1 Celsius) is associated with a decrease of 0.80 units in PT08.S1(CO). However, with p-value = 0.090, this coefficient is not statistically significant at the 5% level.
- **RH (2.32):** A one-unit increase in relative humidity (1%) is associated with an increase of 2.32 units in PT08.S1(CO).
- **AH (13.55):** A one-unit increase in absolute humidity is associated with an increase of 13.55 units in PT08.S1(CO). However, with p-value = 0.082, this coefficient is not statistically significant at the 5% level.

(d) Compare training vs test performance (RMSE and R^2) against SLR.

From Table 8, MLR substantially outperforms SLR on both training and test sets. Train R^2 increases from 0.5817 (SLR) to 0.8389 (MLR), and train RMSE drops from 144.25 to 89.50, confirming that the additional predictors – C6H6(GT), NOx(GT), NO2(GT), T, RH, and AH – capture variation in PT08.S1(CO) beyond what CO(GT) alone explains. Test RMSE also improves from 131.97 to 86.06 and test R^2 from 0.5835 to 0.8229, indicating that adding predictors improves generalization. The small gap between train R^2 (0.8389) and test R^2 (0.8229) suggests the model generalizes well with no signs of overfitting.

Table 8: Train and test performance comparison: SLR vs MLR.

	Model	Train RMSE	Test RMSE	Train R2	Test R2
0	SLR	144.2474	131.9696	0.5817	0.5835
1	MLR	89.5017	86.0606	0.8389	0.8229

(e) Assumption checking and diagnostics

(1) Linearity

To assess linearity, we plot residuals against fitted values and against each of the seven predictors, with a LOWESS smooth (orange line) overlaid to reveal systematic trends. If the linearity assumption holds, residuals should scatter randomly around zero with no systematic pattern.

From Figure 6, the following patterns are observed:

Residuals vs Fitted: The LOWESS line starts near zero but curves downward at higher fitted values, indicating mild non-linearity. The spread of residuals also widens at higher fitted values, suggesting heteroscedasticity. Several extreme negative residuals (below -600) are visible, pointing to influential outliers.

Residuals vs CO(GT): The LOWESS line drifts downward at higher values of CO(GT), suggesting the linear relationship weakens at high concentrations. A vertical band of points around CO(GT) = 6 is visible, corresponding to forward-filled missing values from the data cleaning step.

Residuals vs C6H6(GT): The LOWESS line is flat for most of the range but drops sharply at high C6H6(GT) values (above 40 microg/m3), indicating the linear assumption breaks down at extreme Benzene concentrations.

Residuals vs NOx(GT): Residuals are mostly centered near zero with a flat LOWESS, suggesting reasonable linearity. However, a vertical band around NOx(GT) = 800 and a few extreme negative residuals below -600 are concerning and may reflect outliers or forward-filled values.

Residuals vs NO2(GT): The LOWESS line is nearly flat, suggesting linearity is approximately satisfied. However, residual spread is notably wider at low NO2(GT) values and narrows at higher values, indicating heteroscedasticity.

Residuals vs T: The LOWESS line is flat and close to zero across the full temperature range, suggesting no non-linear relationship. Residual spread is roughly constant, which is the most well-behaved plot among all predictors.

Residuals vs RH: The LOWESS line is flat and near zero, suggesting linearity holds. However, two extreme outliers below -600 and -900 are visible at mid-range RH values, which are worrying and warrant further investigation in the influence diagnostics.

Worrying patterns: The key concerns are (1) the downward-curving LOWESS in the residuals vs fitted plot, indicating the model systematically over-predicts at high sensor responses; (2) the sharp drop in residuals at high C6H6(GT) and CO(GT) values, suggesting non-linearity at extreme concentrations; and (3) several extreme negative residuals (below -600) appearing across multiple plots, pointing to influential outliers that may be distorting the fitted model.

(2) Homoscedasticity

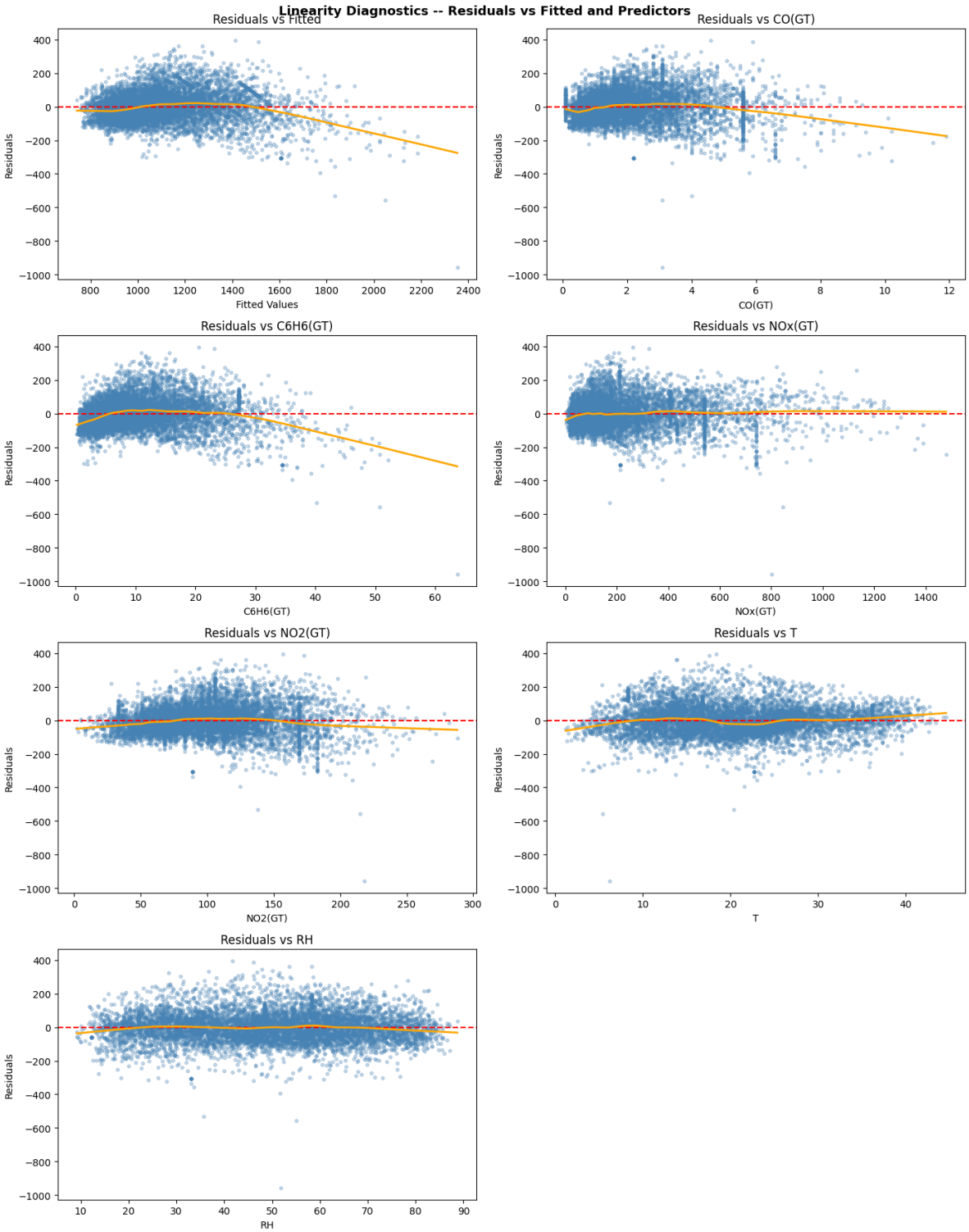

Figure 6: Residuals vs Fitted and Residuals vs each predictor.

To assess homoscedasticity, we plot the square root of absolute standardized residuals against fitted values (Scale-Location plot), with a LOWESS smooth (red line) overlaid. If the constant variance assumption holds, the LOWESS line should be roughly flat and the spread of points should be uniform across all fitted values.

From Figure 7, the LOWESS line starts flat around 0.70 for fitted values between 800 and 1200, then rises steadily to approximately 1.57 at fitted values above 2000. This upward trend indicates that residual spread increases with fitted values – the constant variance assumption is violated. Additionally, several extreme points above 2.0 appear at higher fitted values, further confirming that the model produces larger errors when predicting high PT08.S1(CO) sensor responses. This pattern of heteroscedasticity is consistent with the fanning observed in the residuals vs fitted plot in (a), and is likely driven by the right-skewed distributions of CO(GT) and C6H6(GT) noted in the EDA.

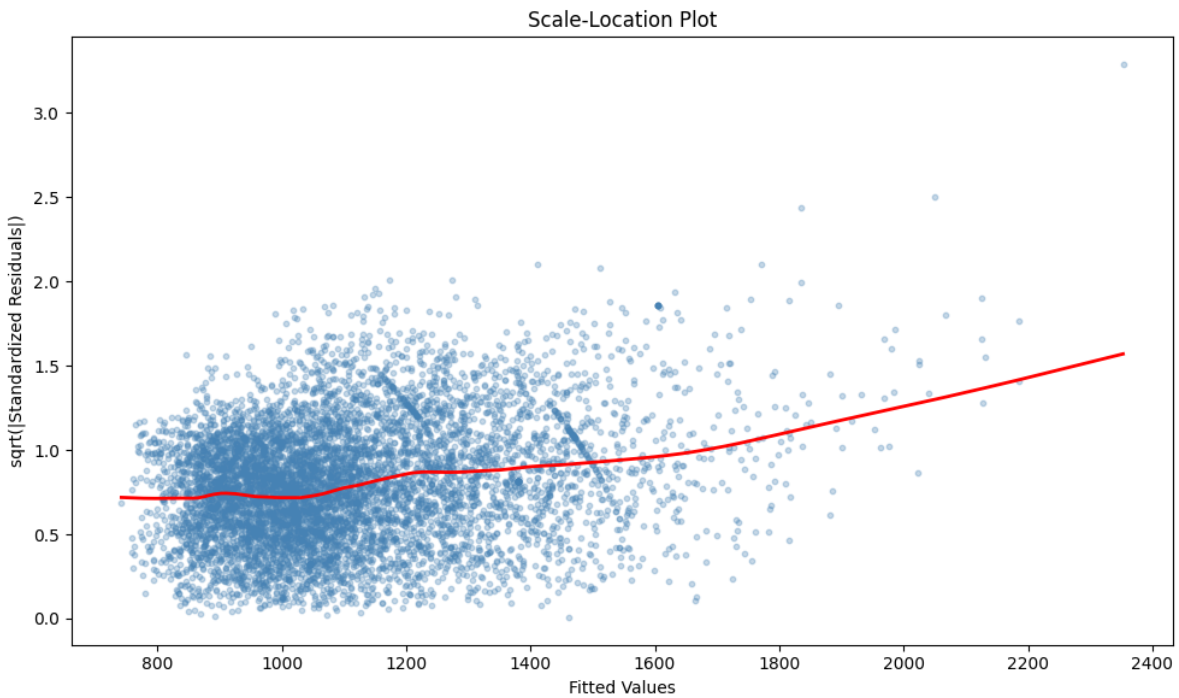

Figure 7: Scale-Location plot for homoscedasticity assessment.

(3) Normality of Errors

To assess normality, we plot the sample quantiles of the residuals against theoretical quantiles from a standard normal distribution (Q-Q plot), with a reference line (red) representing perfect normality. If the normality assumption holds, the points should fall closely along the reference line across the full range.

From Figure 8, the residuals follow the reference line closely in the central region (theoretical

quantiles between -2 and 2), suggesting that the bulk of the errors are approximately normally distributed. However, the plot shows clear departures at both tails. In the upper tail (theoretical quantiles above 2), points curve slightly above the reference line, indicating a mild right heavy tail. In the lower tail, the departures are far more severe – several points fall well below the reference line, with residuals reaching as low as -1000, far beyond what a normal distribution would predict. These extreme negative residuals correspond to the outliers identified in part (a), where the model severely under-predicts PT08.S1(CO) for a small number of observations. Overall, the normality assumption is approximately satisfied in the centre but violated in the lower tail, consistent with the kurtosis of 5.95 reported in the MLR summary.

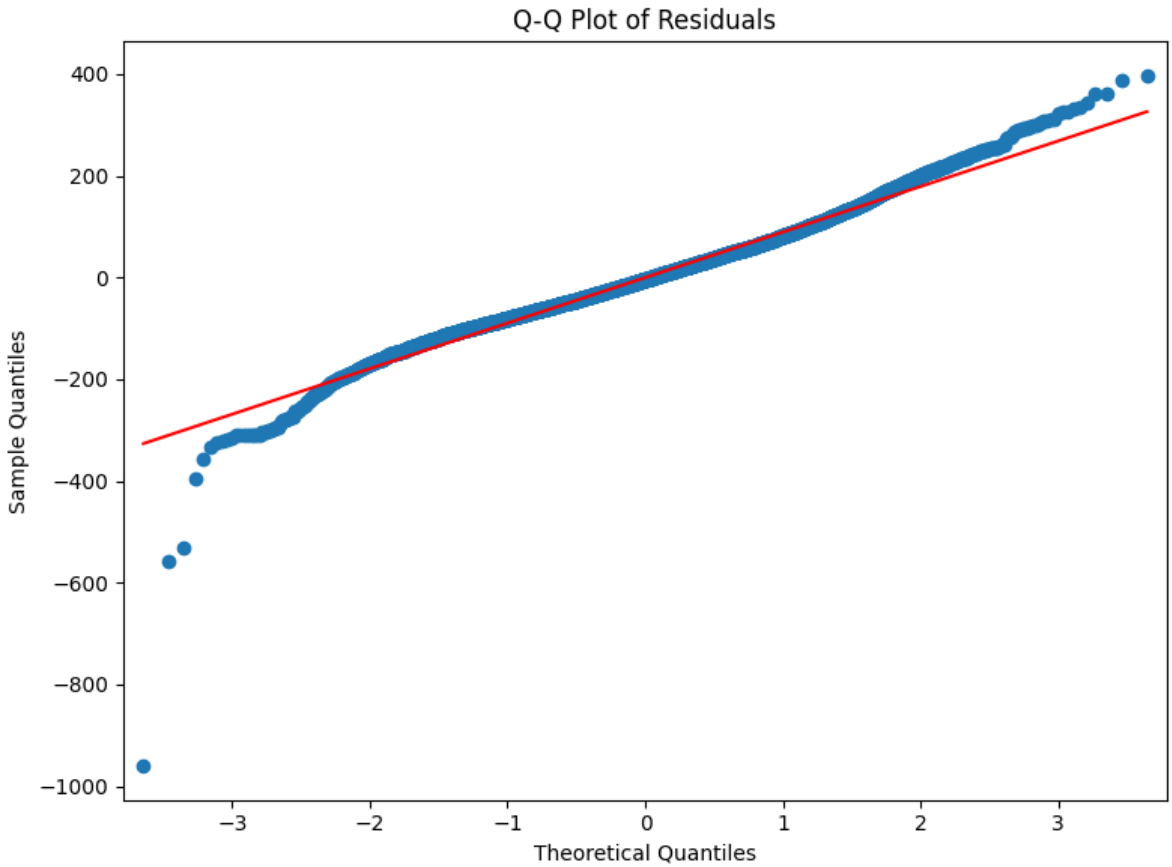

Figure 8: Q-Q plot of MLR residuals.

(d) Outliers and Influence

To identify influential observations, we compute leverage and Cook's distance for each training observation. Leverage measures how far an observation's predictor values are from the mean of all predictors – high leverage points have unusual X values. Cook's distance combines leverage and residual size to measure the overall influence of each observation on the fitted model. The

thresholds used are $2p/n$ for leverage and $4/n$ for Cook's distance, where p is the number of predictors and n is the training sample size.

From Figure 9, the leverage plot shows that most observations fall below the threshold of $2p/n = 0.00187$, but a cluster of points around index 6000-6200 exceeds the threshold, with one observation reaching a leverage of 0.0170. The Cook's distance plot reveals one dominant influential point near index 6100 with Cook's distance of 0.253, far exceeding the threshold of $4/n = 0.00053$, and a few smaller spikes around index 1800 and 4500.

From Table 9, the top 3 most influential observations share a common pattern – all three have large negative residuals (-959, -557, and -532), meaning the model substantially over-predicts PT08.S1(CO) for these hours. The most influential observation (Cook's $D = 0.253$) has an unusually high C6H6(GT) of 63.7 microg/m3 and NOx(GT) of 801 ppb alongside a very low PT08.S1(CO) of 1394, which is inconsistent with what the model expects given those pollutant levels. These observations are physically plausible – they likely correspond to specific pollution events where sensor response did not scale proportionally with ground truth concentrations, possibly due to sensor drift or cross-sensitivity effects documented in the original dataset. While influential, removing them is not recommended without further investigation as they may represent real extreme pollution events.

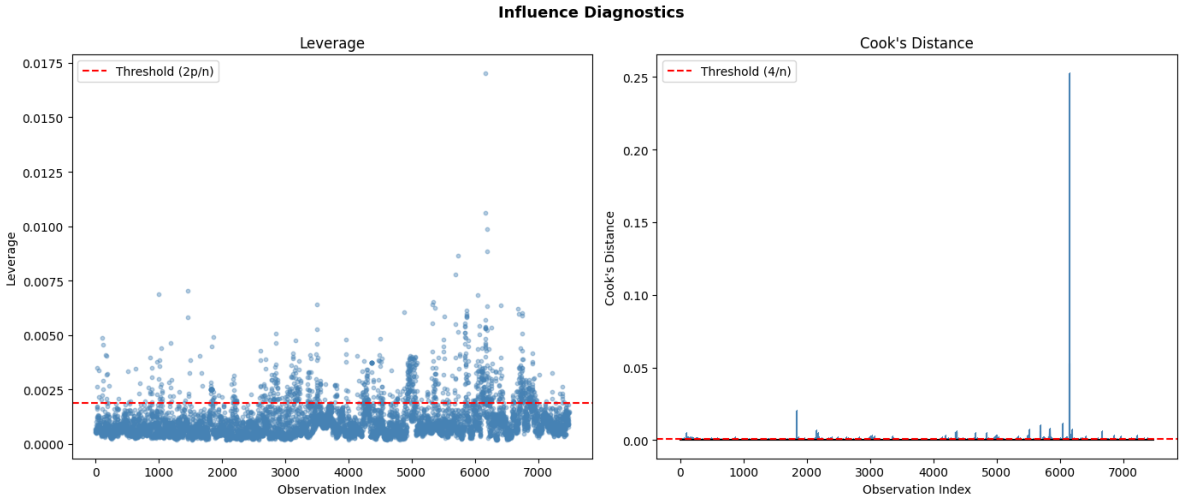

Figure 9: Leverage and Cook's Distance for influential point detection.

Table 9: Top 3 most influential observations by Cook's distance.

	CO(GT)	C6H6(GT)	NOx(GT)	NO2(GT)	T	RH	AH	PT08.S1(CO)	Cook's D	Leverage	Residual
6160	3.1	63.7	801.0	218.0	6.2	51.8	0.4952	1394.0	0.252747	0.017035	-959.0123
6159	3.1	50.7	846.0	215.0	5.4	55.1	0.4970	1493.0	0.052464	0.010630	-556.7119
1839	4.0	40.2	173.0	138.0	20.4	35.7	0.8440	1303.0	0.019984	0.004481	-532.4732

Table 10: Ridge and Lasso regression with 5-fold cross-validation.

```
Ridge selected   : 7488.1039
Lasso selected   : 69.0551
```

9.3 3. Ridge and Lasso Regression

Ridge Regression minimizes the penalized residual sum of squares:

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

Lasso Regression minimizes:

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

where $\lambda \geq 0$ is the tuning parameter controlling the strength of regularization. A larger λ shrinks coefficients more aggressively. Ridge shrinks all coefficients toward zero but retains all predictors, while Lasso can shrink some coefficients to exactly zero, performing automatic variable selection.

(a) To fit Ridge and Lasso regression, predictors are first standardized by subtracting the mean and dividing by the standard deviation, which is required for regularization to treat all predictors equally regardless of their scale. A grid of 200 λ values is searched over the range $[10^{-3}, 10^{6}]$ using 5-fold cross-validation on the training set to select the optimal tuning parameter.

Ridge regression selected $\lambda = 7488.10$, indicating that a strong L2 penalty is needed to shrink the coefficients given the high multicollinearity observed among predictors. Lasso regression selected a much smaller $\lambda = 69.06$, reflecting that the L1 penalty requires less shrinkage to achieve stable estimates while also performing variable selection.

(b) After fitting Ridge and Lasso with the selected λ values, we evaluate cross-validated predictive performance on both training and test sets using RMSE and R^2.

From Table 11, both Ridge and Lasso achieve strong predictive performance. Ridge obtains a test RMSE of 81.29 and test R^2 of 0.8420, while Lasso achieves a test RMSE of 80.19 and test R^2 of 0.8462, making Lasso slightly better on the test set. Both models show a small gap between train and test R^2, indicating good generalization with no overfitting. From Table 12, Lasso shrinks the coefficients of CO(GT), T, and AH exactly to zero, retaining only C6H6(GT), NOx(GT), NO2(GT), and RH as active predictors.

Table 11: Ridge and Lasso cross-validated performance.

	Model	Selected	Train RMSE	Test RMSE	Train R2	Test R2
0	Ridge	7488.1039	90.1285	81.2914	0.8367	0.8420
1	Lasso	69.0551	91.9920	80.1923	0.8299	0.8462

Table 12: Ridge and Lasso estimated coefficients.

	Predictor	Ridge Coef	Lasso Coef
0	CO(GT)	8.3361	0.0000
1	C6H6(GT)	23.2194	22.3539
2	NOx(GT)	-0.1489	-0.0627
3	NO2(GT)	1.0963	1.0481
4	T	0.2642	0.0000
5	RH	2.7159	2.1867
6	AH	0.4325	0.0000

(c) From Table 13 and Figure 10, we compare MLR, Ridge, and Lasso across predictive accuracy, interpretability, and the effect of regularization on collinearity.

Predictive Accuracy: MLR achieves the highest train R^2 (0.8389) but the worst test R^2 (0.8229) and test RMSE (86.06), suggesting mild overfitting due to multicollinearity. Ridge improves test performance with test R^2 of 0.8420 and test RMSE of 81.29, while Lasso achieves the best test R^2 (0.8462) and lowest test RMSE (80.19), indicating that regularization improves generalization beyond what OLS alone can achieve.

Interpretability: MLR retains all 7 predictors with unpenalized coefficients, making it straightforward to interpret. Ridge also retains all 7 predictors but shrinks coefficients toward zero – notably CO(GT) drops from 21.70 (MLR) to 8.34 (Ridge) and AH drops dramatically from 13.55 to 0.43, reflecting their instability under multicollinearity. Lasso is the most interpretable – it shrinks CO(GT), T, and AH exactly to zero, retaining only 4 active predictors: C6H6(GT), NOx(GT), NO2(GT), and RH, identifying these as the most influential predictors of PT08.S1(CO).

Effect of Regularization on Collinearity: The MLR diagnostics flagged a condition number of 4,120 and a large unstable coefficient for AH (13.55, p = 0.082), both indicating multicollinearity. Ridge addresses this by distributing the penalty across all correlated predictors, stabilizing the coefficients without eliminating any. Lasso goes further by zeroing out the most collinear and weakly contributing predictors (CO(GT), T, AH), producing a sparser and more stable model. The improvement in test performance from MLR to Lasso confirms that multicollinearity was inflating variance in the OLS estimates.

Table 13: Comparison of MLR, Ridge, and Lasso regression performance.

	Model	Train RMSE	Test RMSE	Train R2	Test R2
0	MLR	89.5017	86.0606	0.8389	0.8229
1	Ridge	90.1285	81.2914	0.8367	0.8420
2	Lasso	91.9920	80.1923	0.8299	0.8462

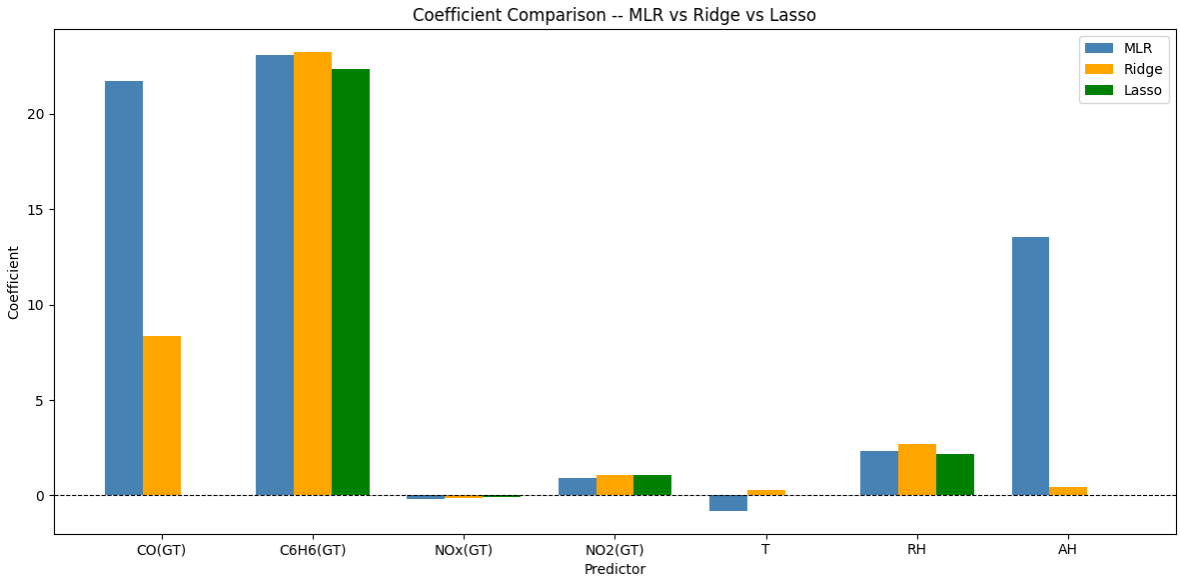

Figure 10: Coefficient Comparison – MLR vs Ridge vs Lasso

10 Result & Model Selecting

Data Summary

After removing NMHC(GT) due to 90.23% missingness and applying forward fill to the remaining missing values, the final dataset retains all 9,357 rows across 7 predictors – CO(GT), C6H6(GT), NOx(GT), NO2(GT), T, RH, and AH – and PT08.S1(CO) as the response variable. The dataset was split 80/20 by time order into a training set (7,485 observations) and a test set (1,872 observations).

Model Performance Summary

From Table 14, all four models show consistent improvement as complexity increases. SLR, using only CO(GT) as a predictor, achieves a test R^2 of 0.5835 and test RMSE of 131.97, explaining only 58% of the variation in PT08.S1(CO). Adding all 7 predictors in MLR substantially

improves performance, with test R^2 rising to 0.8229 and test RMSE dropping to 86.06. Ridge and Lasso further improve generalization through regularization – Ridge achieves test R^2 of 0.8420 and test RMSE of 81.29, while Lasso achieves the best test R^2 of 0.8462 and lowest test RMSE of 80.19 using only 4 predictors.

Assumption Diagnostics Summary

The MLR diagnostics revealed three key violations. First, the residuals vs fitted plot in Figure 6 showed a downward-curving LOWESS trend at high fitted values, indicating mild non-linearity, particularly for CO(GT) and C6H6(GT) at extreme concentrations. Second, the Scale-Location plot in Figure 7 showed an upward-sloping LOWESS trend from 0.70 to 1.57, confirming heteroscedasticity – residual spread increases with fitted values. Third, the Q-Q plot in Figure 8 showed heavy lower-tail departures with residuals reaching as low as -1000, driven by the three influential observations identified in Table 9, which had Cook's distances of 0.2527, 0.0525, and 0.0200 respectively. The MLR summary also flagged a condition number of 4,120, confirming strong multicollinearity among predictors.

Model Selection

Based on Table 14, **Lasso** is selected as the best model. It achieves the highest test R^2 (0.8462) and lowest test RMSE (80.19) among all four models, outperforming MLR by 5.87 RMSE units on the test set. Beyond predictive accuracy, Lasso addresses the multicollinearity flagged in diagnostics by shrinking CO(GT), T, and AH exactly to zero, retaining only 4 active predictors: C6H6(GT), NOx(GT), NO2(GT), and RH, as shown in Figure 10. This produces a sparser and more interpretable model while improving generalization. The small gap between Lasso's train R^2 (0.8299) and test R^2 (0.8462) further confirms stable generalization with no overfitting.

Limitations

Despite strong performance, the selected Lasso model has several limitations. The heteroscedasticity identified in Figure 7 suggests that prediction errors are larger at high sensor responses, meaning the model is less reliable for extreme pollution events. The three influential observations in Table 9, with residuals of -959, -557, and -532, indicate the model struggles with sudden pollution spikes. Additionally, the forward fill applied during data cleaning introduced artificial vertical bands visible in Figure 5 and Figure 6, which may have slightly inflated model performance by reducing true variability in the data.

Table 14: Summary of all model performance metrics.

	Model	Predictors	Train RMSE	Test RMSE	Train R2	Test R2
0	SLR	1	144.2474	131.9696	0.5817	0.5835
1	MLR	7	89.5017	86.0606	0.8389	0.8229
2	Ridge	7	90.1285	81.2914	0.8367	0.8420
3	Lasso	4	91.9920	80.1923	0.8299	0.8462

11 Discussion, Conclusion & Future Work

Discussion

This report set out to investigate whether ground truth pollutant concentrations and environmental variables could reliably predict the PT08.S1(CO) tin oxide sensor response, as a proxy for low-cost sensor calibration in urban air quality monitoring. The results confirm that this is achievable to a high degree – the best model, Lasso, explains 84.62% of the variation in PT08.S1(CO) on unseen test data with a test RMSE of 80.19, using only 4 predictors: C6H6(GT), NOx(GT), NO2(GT), and RH. This is consistent with the physical understanding of the sensor – Benzene (C6H6) is the strongest individual predictor (r = 0.886), reflecting the known cross-sensitivity of tin oxide sensors to aromatic hydrocarbons documented in De Vito et al. (2008). The fact that CO(GT) – the pollutant the sensor nominally targets – was zeroed out by Lasso is a notable finding, suggesting that in the presence of correlated pollutants, Benzene concentration is a more stable and informative predictor of this sensor's response than CO itself. This highlights the cross-sensitivity problem at the core of metal oxide sensor calibration and demonstrates the value of regularization methods in identifying the true drivers of sensor behavior.

However, several important limitations were uncovered. The assumption diagnostics revealed clear heteroscedasticity – residual spread increases with fitted values – meaning the model is less reliable at predicting high sensor responses corresponding to extreme pollution events. The Q-Q plot showed heavy lower-tail departures with residuals reaching as low as -1000, driven by a small number of highly influential observations identified through Cook's distance analysis. These observations likely correspond to sudden pollution spikes or sensor anomalies that the linear model cannot capture. Furthermore, the forward fill applied during data cleaning, while appropriate for a time-series dataset, introduced artificial continuity around missing value blocks, which may have slightly inflated model performance by reducing true variability. What this analysis did not explore includes temporal patterns – hour of day and seasonal effects – which the EDA suggested are present in the data through the bimodal distribution of temperature and the time-series structure of pollutant concentrations.

Conclusion

This report demonstrated that linear regression methods, particularly Lasso regression, can effectively calibrate metal oxide sensor responses using ground truth pollutant concentrations and environmental variables. Starting from a baseline SLR model with test R^2 of 0.5835, performance improved substantially through MLR (0.8229), Ridge (0.8420), and finally Lasso (0.8462), with each step motivated by the multicollinearity and assumption violations identified in diagnostics. Lasso is selected as the final model for its superior predictive accuracy, interpretability, and automatic variable selection, reducing the predictor set from 7 to 4 while improving generalization. The analysis confirms that cheap sensor responses can be reliably estimated from reference measurements, supporting the broader goal of scalable low-cost air quality monitoring in urban environments.

Future Work

Several directions could extend this analysis. First, non-linear models such as Random Forest or Gradient Boosting could address the heteroscedasticity and non-linearity observed in diagnostics, likely improving performance at extreme pollution levels. Second, temporal features such as hour of day, day of week, and month could be engineered from the datetime index to capture diurnal and seasonal patterns not exploited in the current models. Third, the three remaining sensor responses – PT08.S2(NMHC), PT08.S3(NOx), PT08.S4(NO2), and PT08.S5(O3) – could be modeled using the same pipeline to provide a complete multi-output sensor calibration framework. Finally, robust regression methods or log transformations of skewed predictors could be explored to address the heavy-tailed residuals and influential observations identified in the Cook's distance analysis.

12 References

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13 Appendix

All code, data, and outputs for this project are available at:

https://github.com/3313556500/210P-Final-Project—Tao-Chen.git